

Week 2

23 – 27 June

Goals

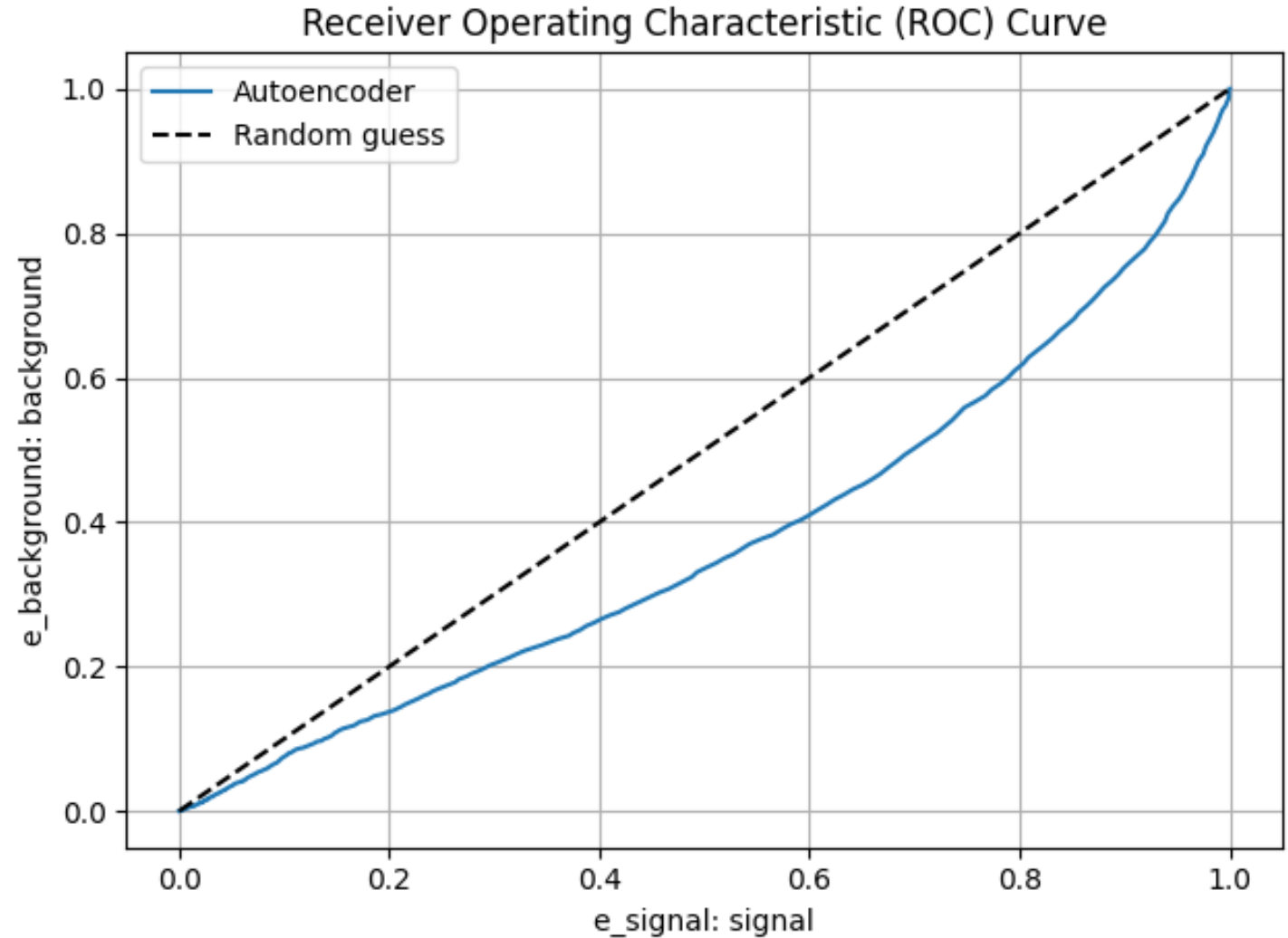
- Optimize parameters in graph model
- Parameters:
 - Learning rate
 - Nearest neighbours
 - Smallest dimension (latent space)
- Determine best parameter values based on AUC score

Original Config

- Learning rate: 0.00005
- Smallest dim: 16
- Nearest neighbours: 16

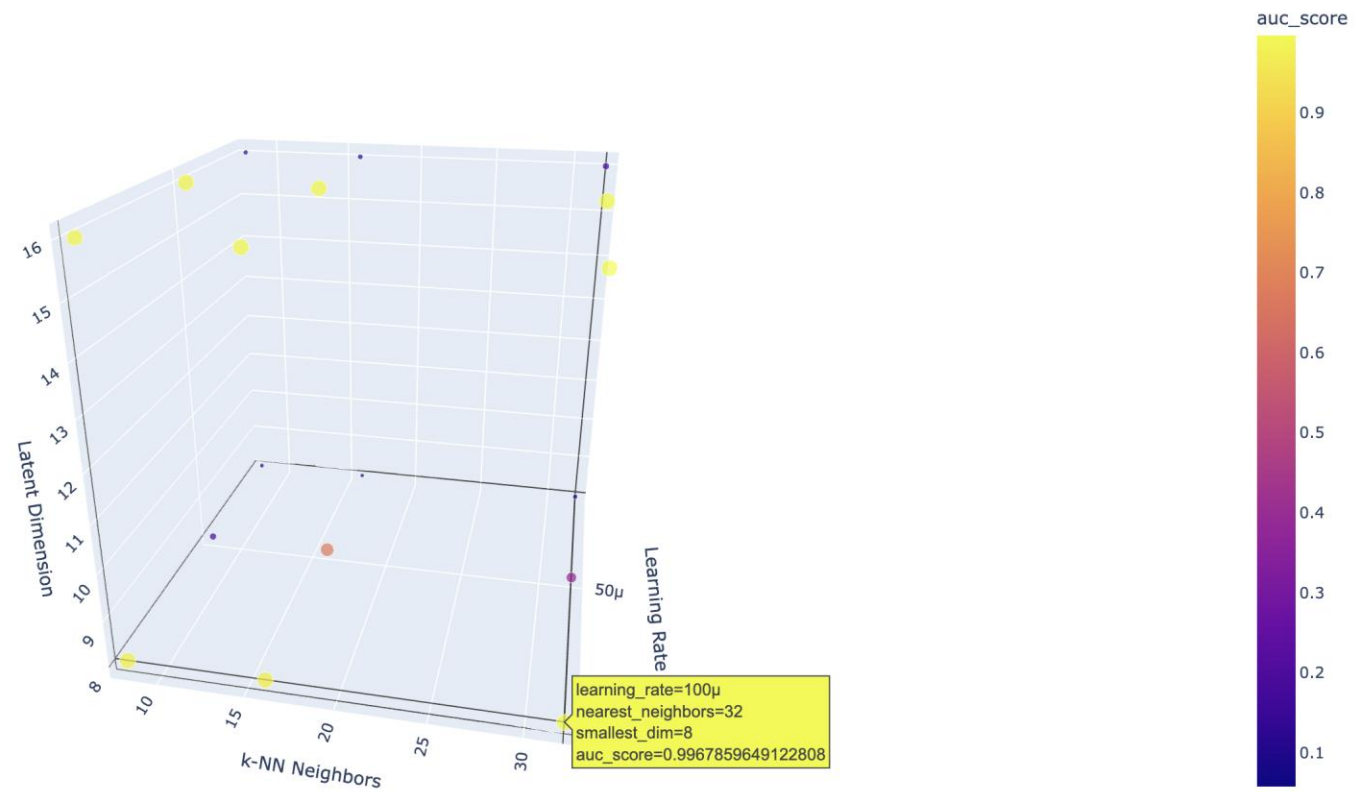
Original Config ROC

- AUC = 0.621



3D Scatterplot

3D AUC Scatter (weight_decay=0.0001)

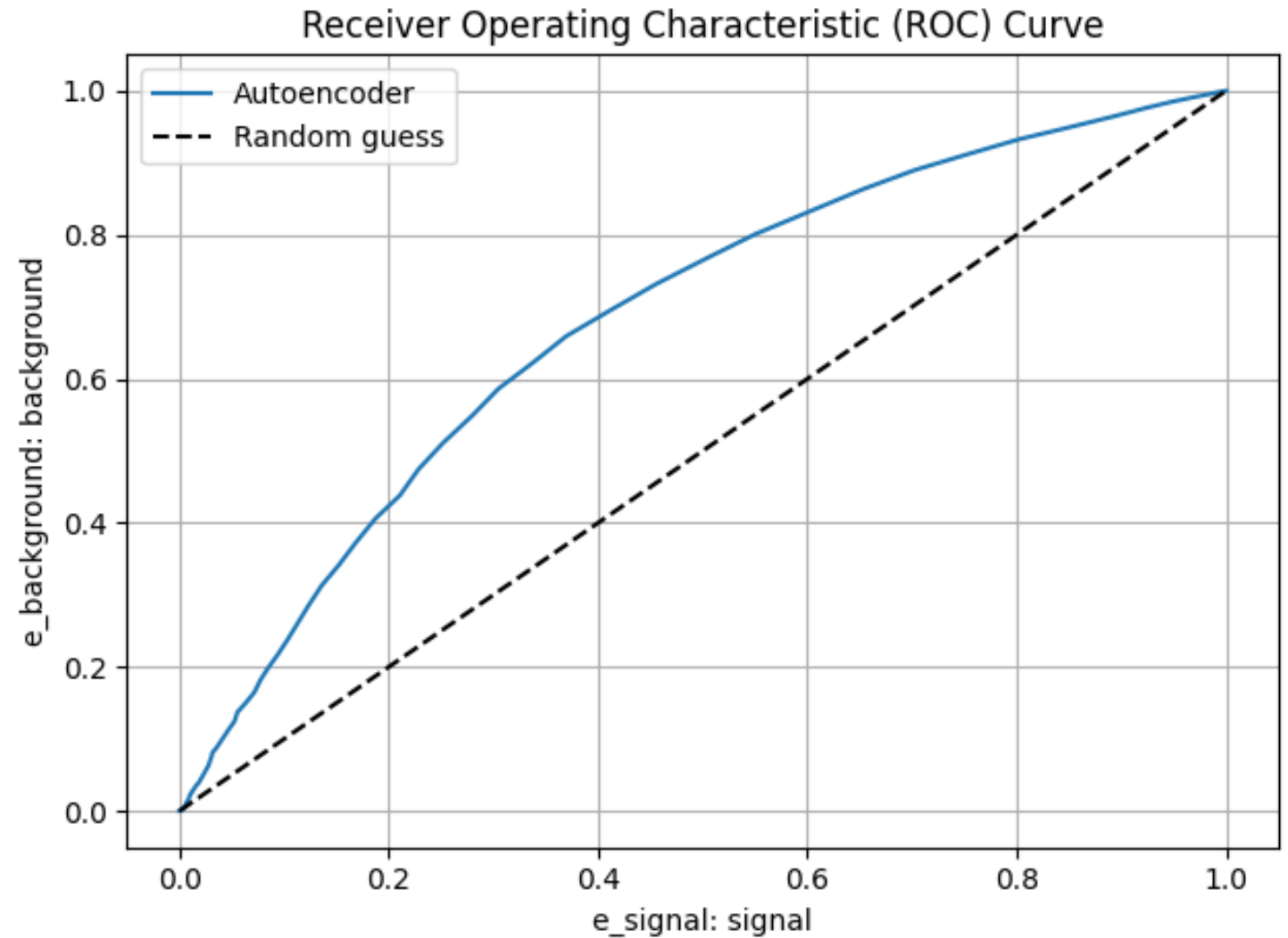


New Config

- Learning rate: 0.0001
- Smallest dim: 8
- Nearest neighbours: 32

New Config ROC

- AUC = 0.318



Next Steps

- Try different jets for signals, other than WJets
- Change edges to other values rather than Euclidean distance
 - Invariant mass